



IMT Atlantique

Bretagne-Pays de la Loire

École Mines-Télécom

Ocean warming contribution to Sea level rise

UE Big data & Cloud computing

Project 6

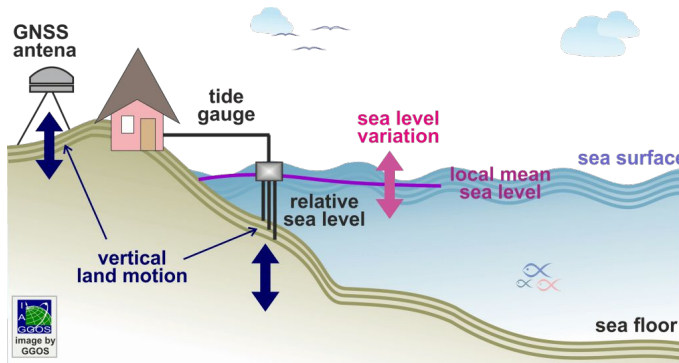
Nizar Zerrad
Mouad Obeidi
Ghaly Sentissi El Idrissi
Martin Coubard



Context

In situ and satellites measurements have shown that sea is rising:
1.7mm/year since 20th century

In situ → tide gauge



Sea level monitoring with tide gauges, ©GGOS

Observation from space



- Sea Surface Temperature, Salinity, ...
- Quasi-totality of the ocean
- Measures every few days

Complete coverage of satellite revealed that the rise in sea level is not uniform !

What is impacting sea level rise ?

Global mean sea level provides a measure of the state of the climate system, driven by 3 processes :

Land water storage

Water removed from land (groundwater pumping, dam building, ...)

Ice Melt

Ice sheets and mountain glaciers are melting => add fresh water to the ocean

Steric effect

Caused by changes in the ocean density, from temperature or salinity

Steric sea level change

TSLA + HSLA

Thermosteric Sea Level Anomaly (or *thermal expansion*): **TSLA**

Increasing temperature reduces ocean density and increases the volume per unit of mass => increasing sea level.

Halosteric Sea Level Anomaly: **HSLA**

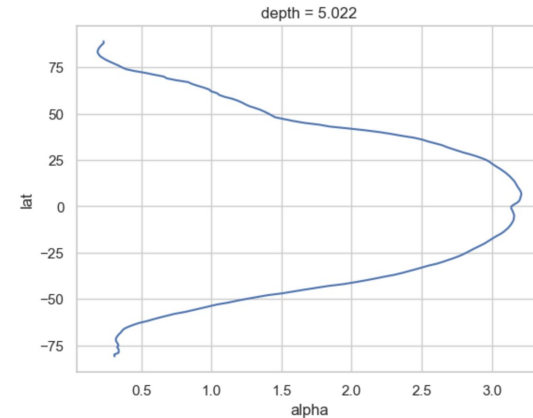
Higher salinity leads to higher density and decreases the volume per unit of mass => reducing sea level.

Temperature and Salinity anomalies

$$TSLA = \int_{-H}^0 \alpha \Delta T dz$$

$$HSLA = - \int_{-H}^0 \beta \Delta S dz$$

α : ocean thermal expansion coefficient
 β : haline contraction coefficient



Aim of the study: Analyze the steric contribution to Sea Level Anomaly trend

Dataset presentation

First dataset

We have a worldwide dataset of sea surface height (SSH) measurements **from satellites** with a resolution of 1° .

Variables :

SSH	- 83 < latitude < 90	1 < longitude < 360	time per month between 1993 and 2018
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Size of the dataset : 18Gb

Utility: We will use it in order to compute local sea surface anomalies

Dataset presentation

Second dataset

EN4 dataset from in situ measurements :

It combines the profile data to create global maps of temperature and salinity. These maps also include uncertainty estimates, which show how confident we can be about the data in different areas.

Resolution of 1°

Variables :

$5 \text{ m} < \text{Depth} < 5350 \text{ m}$	$- 83^\circ < \text{Latitude} < 90^\circ$	$1^\circ < \text{Longitude} < 360^\circ$
Salinity	Temperature	Time per month between 1993 and 2010



Argo sensor, © LOPS

Size of the dataset : 52Gb

Utility: We will use it in order to compute Thermosteric and Halosteric effects

Coiled presentation



Coiled is a platform that provides managed solutions for Dask that is used for parallel computing. It simplifies scaling Python workloads in the cloud, making it easier for users to work with large datasets and perform distributed computations.

- **Managed Dask Clusters:**
 - Coiled lets you easily create, manage, and scale Dask clusters in the cloud (AWS, Azure, GCP).
 - You don't need to set up or maintain the infrastructure manually.
- **Cost Management:** Provides insights into resource usage and costs, helping you optimize cloud expenses.
- **Security:** Provides secure integrations with cloud environments, ensuring sensitive data stays safe.

Workspace: **class-2025**

RUNNING CLUSTERS								
Name	Creator	Status	Workers	Uptime	Approx Clo...	Used Credits	Tasks	Flags
notebook-user	louiscbns	ready	0/0	8m 14s	\$0.12/hr (\$0.0 2/hr (0.27 so f		0	
notebook-zeineb	gmaze	ready	0/0	2h 20m	\$0.24/hr (\$0.5 4/hr (9.38 so f		0	
notebook-p5	gmaze	ready	0/0	2h 30m	\$0.24/hr (\$0.5 4/hr (10.04 so		0	
ds2-project5	gmaze	ready	2/5	2h 30m	\$0.44/hr (\$1.5 34/hr (128.37 €		76 336	🔍 ⚙️ 🔄
ds2-project6	gmaze	ready	6/6	2h 48m	\$1.08/hr (\$3.0 98/hr (275.48		18 346	🔍 ⚙️ 🔄

Rows per page: 10
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How to use Coiled ?



```
import coiled
cluster = coiled.Cluster(name="ds2-highcpu-binder", workspace="class-2025")
# cluster = coiled.Cluster(name="ds2-highmem-binder", workspace="class-2025")
client = cluster.get_client()
```

Coiled Cluster

<https://cloud.coiled.io/clusters/734119?account=class-2025>

Overview

Name: ds2-highcpu-binder
Env: ds2-coiled-2025-binder
Scheduler Status: started
Dashboard:
<https://cluster-dtgpe.dask.host?token>

Configuration

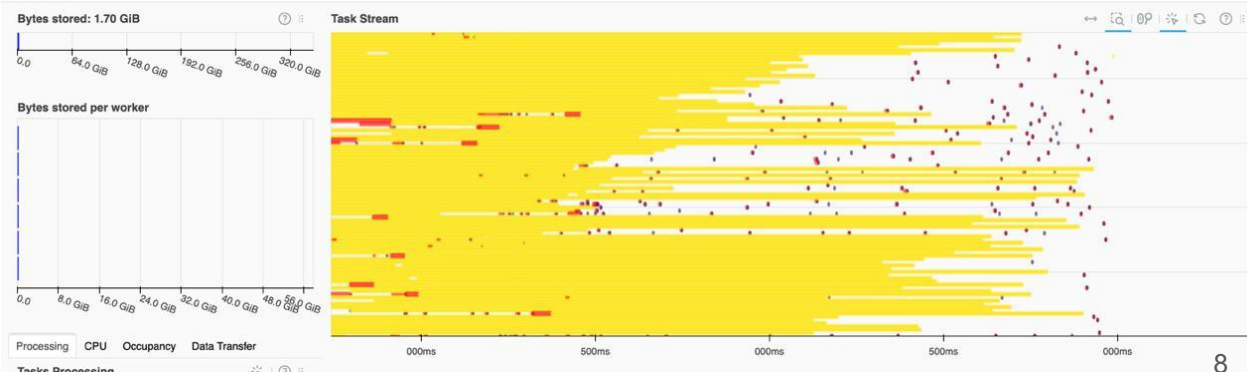
Region: us-east1
Scheduler: n1-highmem-2
Workers: n1-standard-16 (10)
Workers Requested: 10

(2025/01/21 10:55:19 CET)

Most of your workers have arrived. Cluster ready for use.

Integration with Python Ecosystem: Works seamlessly with Python libraries like Pandas, NumPy, and Scikit-learn for large-scale computation.

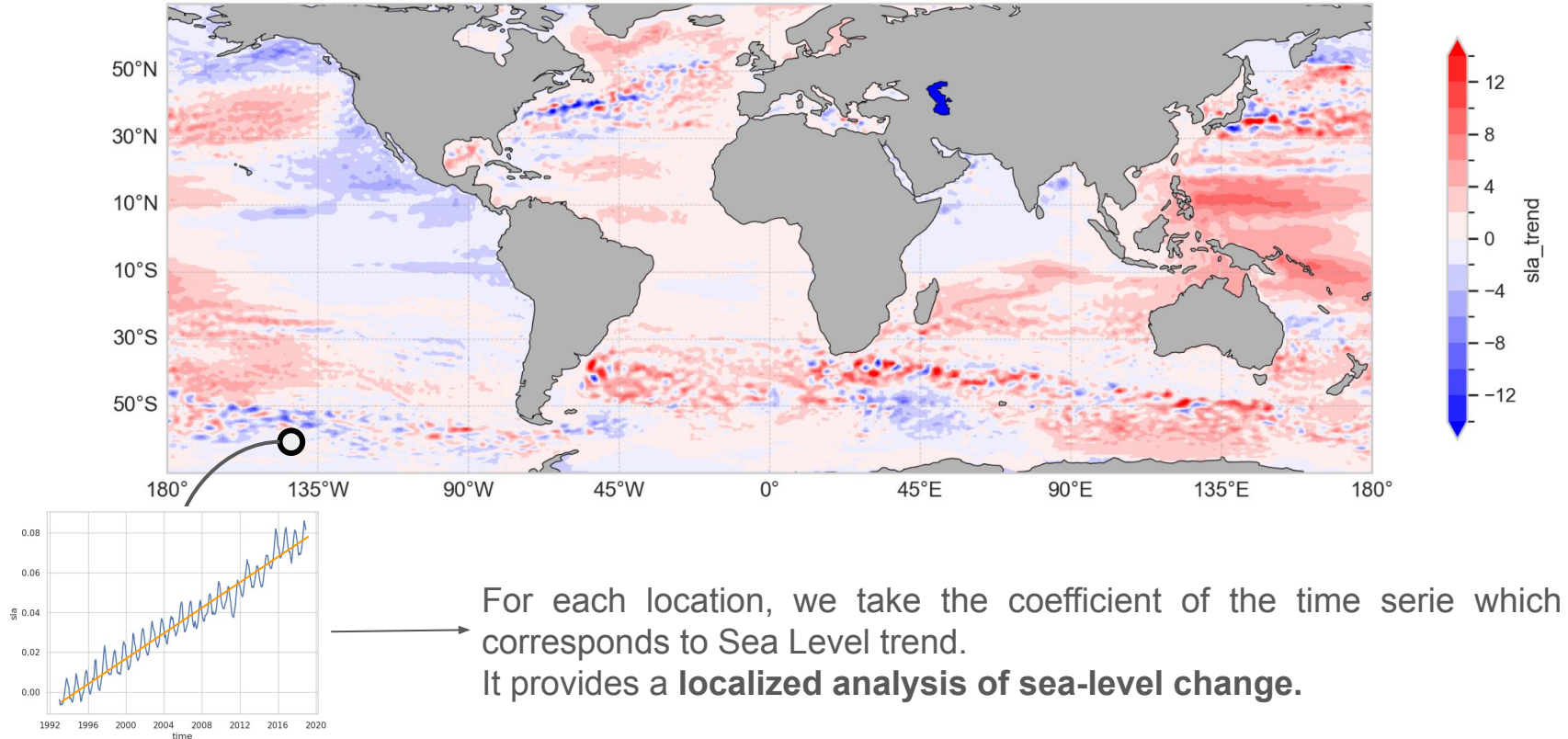
User-Friendly Interface : We can see in the dashboard the real time activity of the cluster



Results presentation

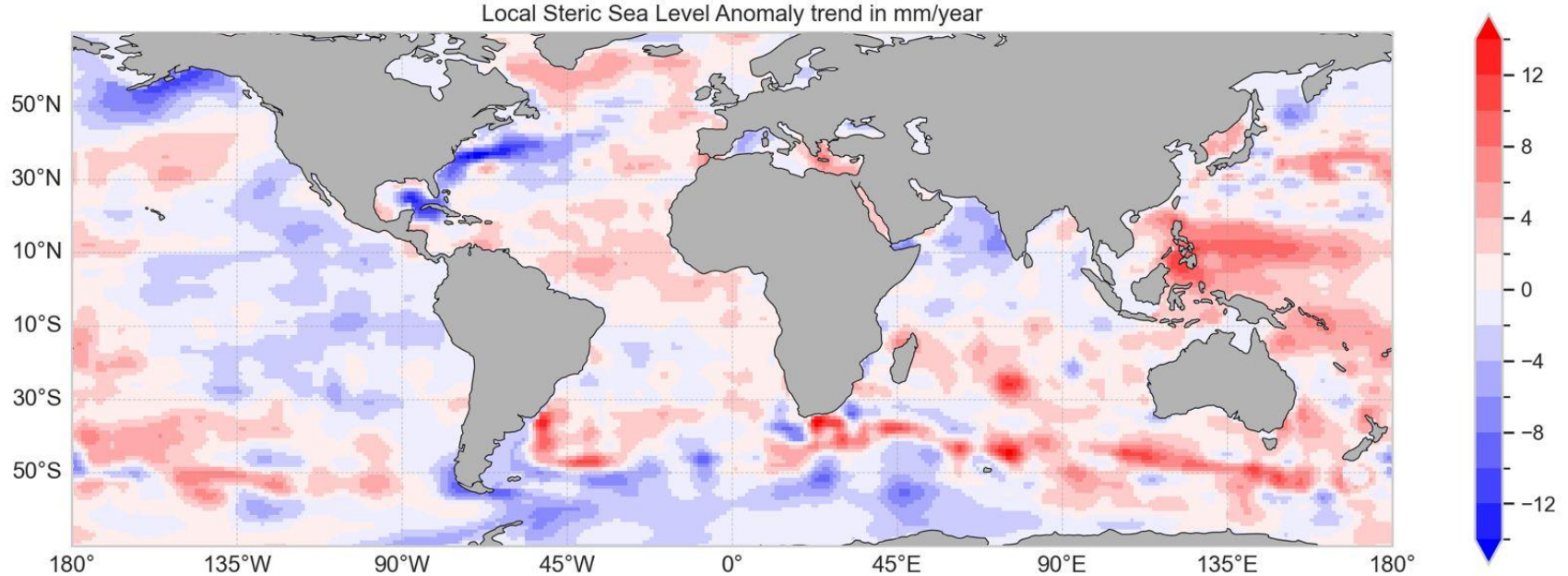
Sea level anomaly trend from satellite dataset (1st one)

Local Sea Level Anomaly trend in mm/year



Results presentation

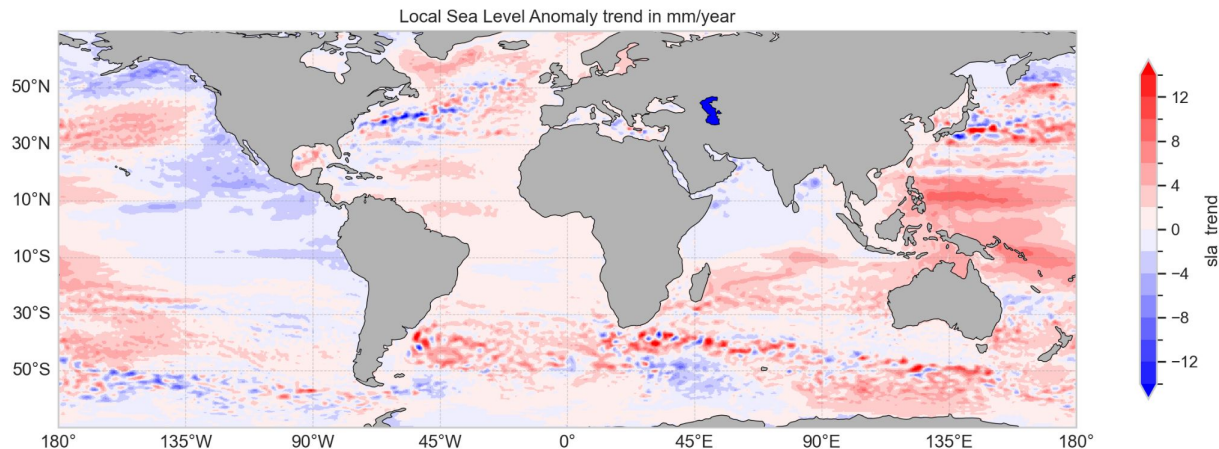
Local Steric Sea Level Anomaly trend from in situ dataset (2nd one)



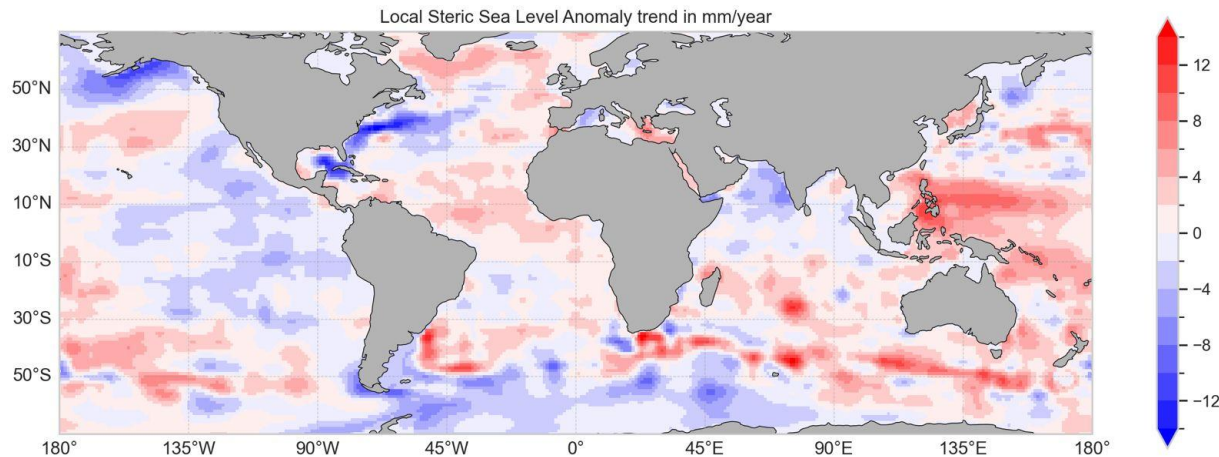
$$TSLA = \int_{-H}^0 \alpha \Delta T dz \quad + \quad HSLA = - \int_{-H}^0 \beta \Delta S dz$$

Results comparison

First dataset



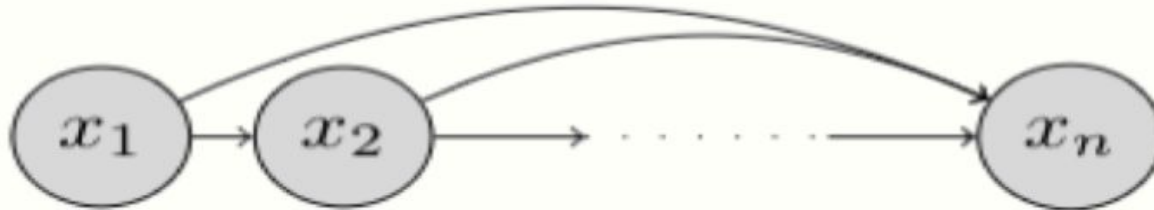
Second dataset



Choice of model

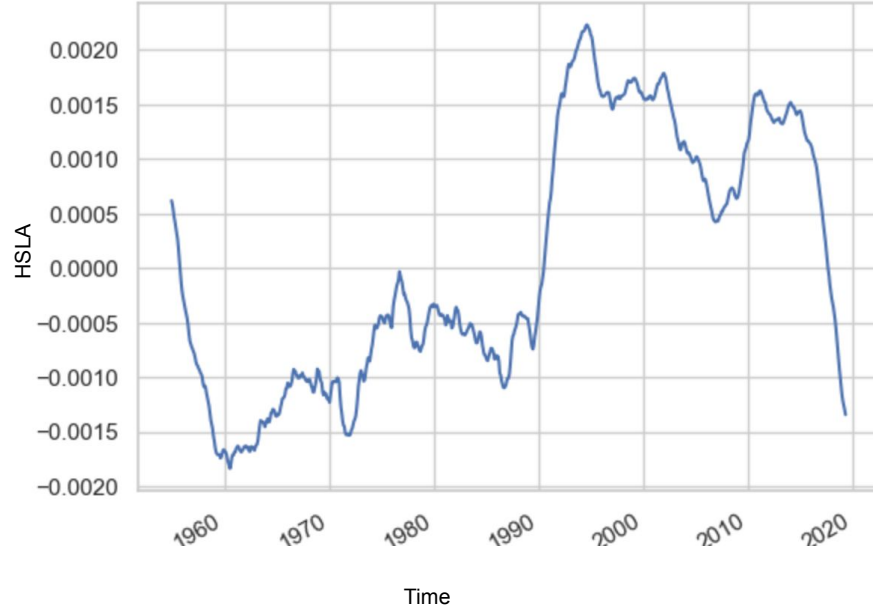
according to our knowledge we choose the autoregressive model:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + \varepsilon_t$$



Times series

Mean HSLA over months

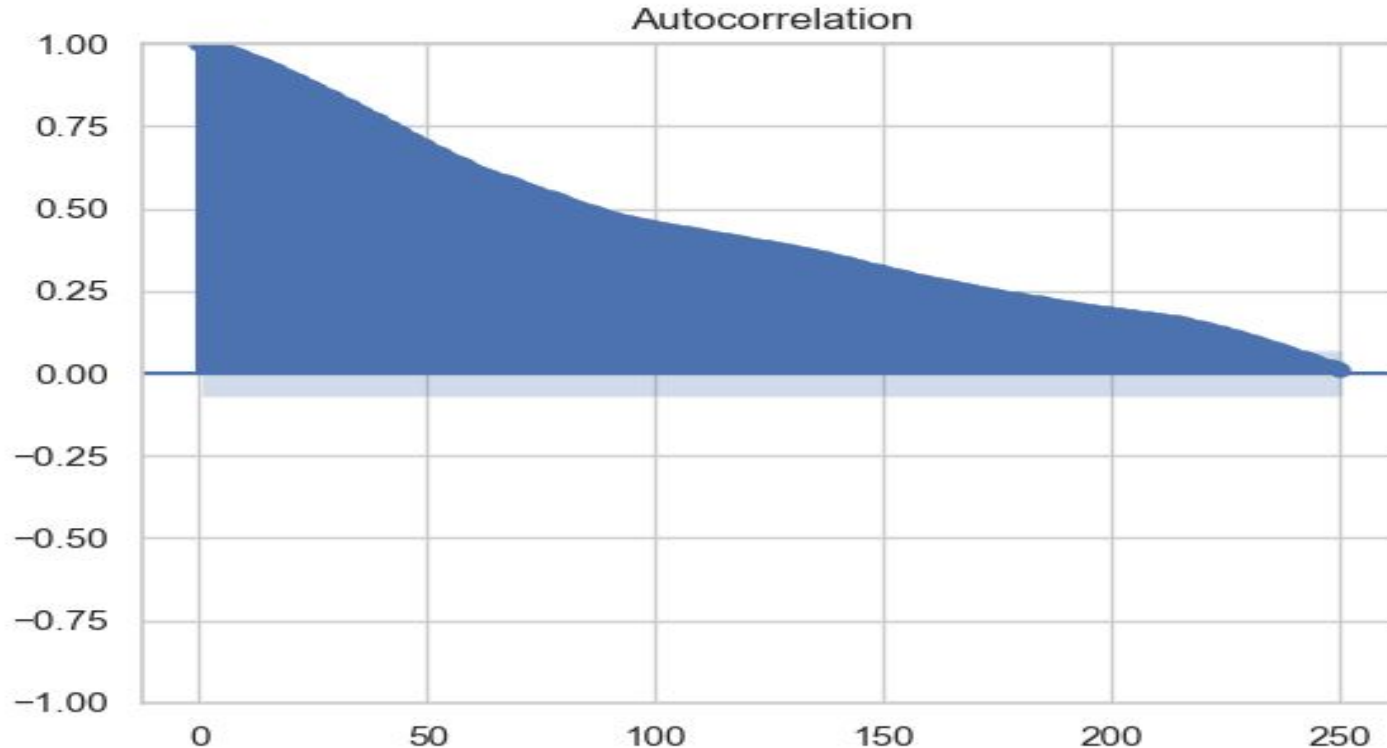


Mean TSLA over months



Choice of parameters

We plot the autocorrelation function to know what would be a reasonable choice of lags

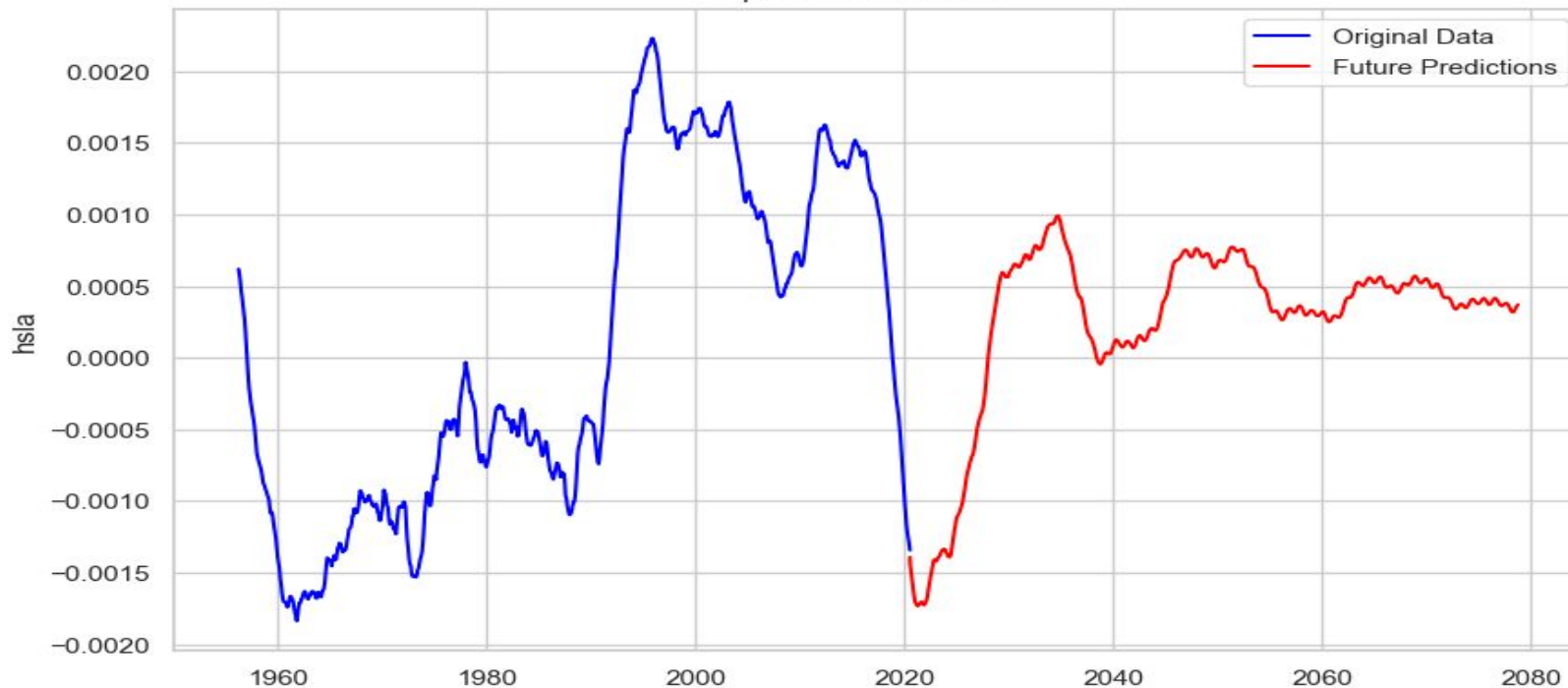


Prediction on HSLA

Halosteric Sea Level Anomaly

mean HSLA over lat lon weighted by surface elements

prediction until 2080

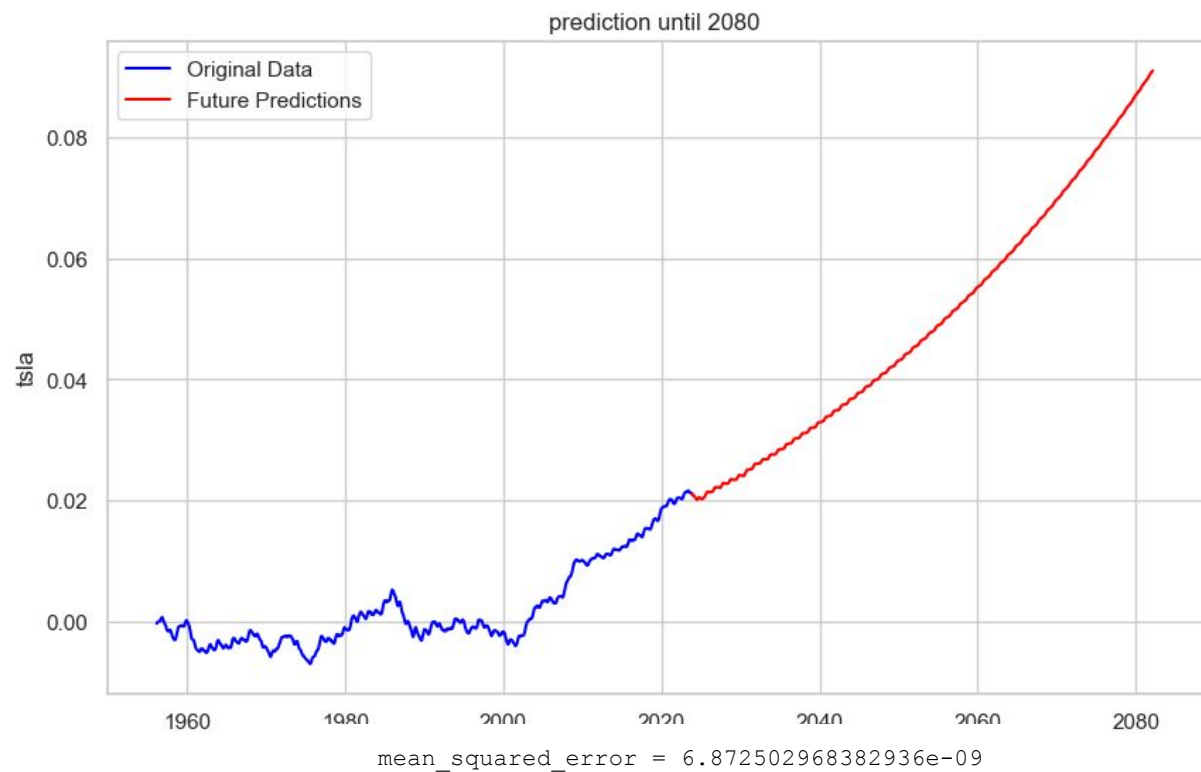


`mean_squared_residuals = 1.540588940093672e-10`

Prediction on TSLA

Thermosteric Sea Level Anomaly

mean TSLA over lat lon weighted by surface elements



- **Modeling Errors:** estimate up to when could our predictions be reliable
- **perform a better model :** ARIMA model
- **Sensitivity analyses:** Perform sensitivity analyses to assess how assumptions about ocean density changes affect the results.
- **Using more complexe models:** Leveraging (LLMs) . LLMs fine tuned on time series needs more computing resources (GPUs).
- **Use Cloud to make our model autonomous:** apply automated Continuous training using version control.

Thank you for your attention !